

CSC-251: Database System (4: 3+1)

General Information

Course Number	CSC-251
Credit Hours	3+1 (Theory Credit Hour = 3, Lab Credit Hours = 1)
Prerequisite	None

Course Objectives

The aim of this course is to build the foundation of database design, implementation and management from theory and practical perspectives. The focus of this course would be on design and development of relational model and relational query languages associated with such model. The basic concepts of concurrency control and transaction processing will also be covered. In addition to the theoretical concepts, the course through the lab section, and programming assignments and projects, will require students to use the Oracle / MySQL database systems and develop term projects.

Catalog Description

CSC-251

Course Content

Session No.	Week No.	Topic	Suggested Readings (Chapters)
01-05	1-2	Introduction to DBMS - Database-System Applications - Views of Data - Database Languages - Relational Databases - Database Design - Data Storage and Querying - Transaction Management - Database Architecture - Database Users and Administrators	1
06-10	2-3	Introduction to the Relational Model - Structure of Relational Databases - Database Schema - Keys - Schema Diagrams - Relational Query Languages - Relational Operations	2
11-14	4-5	Relational Model - Relational Operations - Formal Relational Query Language - Relational Algebra	2.6
14-18	5-6	An algebraic Query Language - Natural Joins - Theta- Joins - Combining operations to form queries - Naming and renaming - Relationships among operations	2.6
18-23	7-8	Functional-Dependency Theory - Functional Dependencies (FDs) - Definition of FD	7

		• Keys of relations • Super Keys • Rules about FDs • Reasoning about FDs • Armstrong’s Axiom and inferencing Rule • Computing the closure of Attributes • Computing the closure of FDs	
Mid Exam			
24-26	9	Design of relational database schema • Anomalies • Decomposing Relations using FDs • Third Normal Form (3NF) and Boyce-Codd Normal Form (BCNF)	7
27-29	10	Overview of the Design Process • The Entity-Relationship Model • Constraints • Removing Redundant Attributes in Entity Sets • Entity-Relationship Diagrams • Reduction to Relational Schemas • Entity-Relationship Design Issues • Extended E-R Features	6
30-33	11	Enhanced ER design • From ERD to Relational Designs (translation) • From Entity sets to relations • From ER Relationships to Relations • Combining relations	6
34-39	12- 13	Transaction Management • ACID Properties • Concurrency Control • Concurrency Control with locking methods • 2PL • Deadlock	17, 18
40-42	14	Semi-Structured and Un-structured Data • XML • JSON • PL/SQL • NoSQL • Graph Databases	11
43-45	15	Project Presentations and Revision	
Final Exam			

Text Book

Avi Silberschatz, Henry F. Korth, S. Sudarshan: "Database System Concepts", 7th Edition, 2019, McGraw-Hill

Reference Material

Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom:, "Database Systems: The Complete Book ", 3rd edition, Pearson Prentice Hall

Thomas Connolly, Carolyn Begg:, "Database Systems: A practical approach to design, implementation and Management ", 5th edition, Addison Wesley

Course Learning Outcomes

	Course Learning Outcomes (CLO)
1	Explain fundamental database concepts
2	Design conceptual, logical and physical database schemas using different data models
3	Identify functional dependencies and resolve database anomalies by normalizing database tables
4	Use Structured Query Language (SQL) for database definition and manipulation in any DBMS

CLO-GA Mapping

CLOs	GA1	GA2	GA3	GA4	GA5	GA6	GA7	GA8	GA9	GA10
CLO 1	1									
CLO 2		x								
CLO 3				x						
CLO 4					x					

Computer Science/Software Engineering Program Learning Outcomes

GA: Graduate Attributes

GA-1 Academic Education: Completion of an accredited program of study designed to prepare graduates as computing professionals.

GA-2 Knowledge for Solving Computing Problems: Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.

GA-3 Problem Analysis: Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

GA-4 Design/Development of Solutions: Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs.

GA-5 Modern Tool Usage: Create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

GA-6 Individual and Teamwork: Function effectively as an individual and as a member or leader of a team in multidisciplinary settings.

GA-7 Communication: Communicate effectively with the computing community about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

GA-8 Computing Professionalism and Society: Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the responsibilities relevant to professional computing practice

GA-9 Ethics: Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

GA-10 Life-long Learning: Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.